

Voice Cloning using RVC: A Lightweight and Personalized Speech Synthesis Technique

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Abstract—This study introduces a lightweight and efficient voice cloning system using Retraining-based Voice Conversion (RVC) to deliver individual speech synthesis with the help of a small amount of target speaker data. The proposed approach retains the linguistic content and converts the speech of the source speaker into the voice features of the target speaker. Quantitative evaluation is carried out with Mel Frequency Cepstral Coefficient (MFCC) cosine similarity. The scores of 0.71 and 0.9935 between the source and target speech and synthetic and target speech, respectively, denote strong acoustic alignment. Moreover, the speech generated has temporal and spectral coherence when qualitatively analyzed using the parameters of waveform, spectrogram, pitch contour, and chroma. The system can be used in applications such as voice personalization, assistive speech technology, and low-resource speech synthesis, where the real-time inference property of the system is important.

Index Terms—Voice Cloning, RVC, Speech Synthesis, MFCC, HuBERT, FAISS

I. INTRODUCTION

Developments in deep neural network architecture and machine learning have resulted in a rapid increase in the synthesis of human-like speech becoming an important component of modern human-computer interaction. The potential of machines to generate natural, expressive, and customized audio has wide applications in assistive technology, education, entertainment and accessibility. One of the simplest forms of communication is speech. In the past several decades, speech synthesis has evolved not only out of concatenative and parametric techniques but also out of data-driven neural models that are capable of synthesizing speech that sounds natural and has a high level of comprehensibility [9].

Although there have been these advancements, personalization continues to be a major challenge in modern speech synthesis systems. Several commercial and research-based solutions rely on canned voice templates or require a substantial quantity of high-quality and speaker-specific recordings to adapt to a new voice. These conditions limit access to individuals and personalization is costly and not possible when there are limited resources. Additionally, the existing systems often find it hard to preserve such expressive characteristics as prosody, speaking style, emotional nuances, and the speech turns mechanical and unnatural.

To address these shortcomings, this paper proposes a streamlined and customized voice cloning system, which is

designed on RVC. The suggested method allows for speaker adaptation with just a limited amount of target speaker data, and makes it perfect in cases when big recordings are inaccessible. Once the system has been trained, it converts the source speech into the synthetic audio that retains the original linguistic content but assumes the acoustic characteristics of the target speaker, such as pitch changes, tone color and speech rhythm. The design is efficiency-based and real-time inference that allows operation in low-resource environments. [10]. The pipeline based on the RVC is based on the self-supervised representation learning, trusted pitch extraction, and GAN-based vocoding to build a modular and scalable pipeline. This modularity will be easily adjusted with minimal data preparation and reduces the need to rely on manual transcription or large annotated datasets. As a result, the system is easily adaptable to different speakers and usage conditions at very low computational cost.

The possibilities of the personalized voice cloning are wide. In assistive technology, the systems are used to enable individuals who have speech impairments to have their voices restored or speech personalized. Lightweight voice cloning can facilitate dubbing, narration, and voice over production in the media production and entertainment. Speech synthesis can be customizable and expressive in educational tools, interactive storytelling and virtual agents. [11]. This research contributes to the existing efforts to create more accessible and practical personalized speech synthesis systems by exploring the use of RVC in low-data voice cloning and offering objective and qualitative assessments of its application. The paper also highlights the current constraints and recommends the way forward in terms of assessing and responsible use of voice cloning technologies in the future. The inference flow of the proposed work is shown in Fig. 1.

Although neural speech synthesis has advanced very fast, the current state of art systems tend to need large amounts of high quality labeled data and substantial amounts of computational power, rendering them unfeasible in the real world personalization setting. In many applications, such as assistive technologies or low resource environments, only a small amount of speaker specific data is available. Therefore, there is a critical need for lightweight and efficient voice cloning systems that can deliver high quality and speaker consistent speech with minimal data and computational overhead.

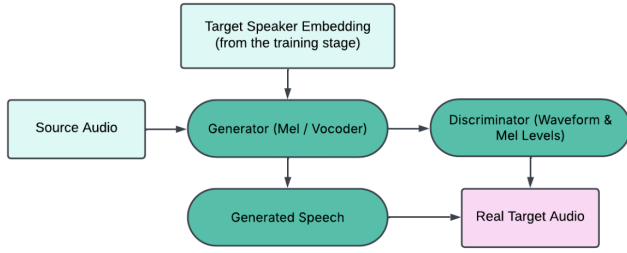


Fig. 1. The generator takes as input the source speech along with the target speaker’s embedding to produce a synthetic audio waveform. This output is evaluated by a discriminator trained to distinguish between real and generated speech. The adversarial setup enables the generator to iteratively improve audio realism by learning from discriminator feedback, resulting in high quality, expressive, and speaker consistent voice outputs.

Key Contributions:

- Developed a lightweight RVC-based voice cloning framework capable of adapting to a target speaker using limited data.
- Integrated HuBERT, RMVPE, FAISS, and a GAN-based vocoder for efficient and high-quality speech synthesis.
- Enabled real-time inference suitable for deployment in resource-constrained environments.
- Designed a combined evaluation framework using both objective metrics (MFCC cosine similarity) and qualitative analyses (spectrogram, pitch, chroma, waveform).

II. LITERATURE REVIEW

Voice cloning and neural speech synthesis have witnessed rapid progress in recent years, driven by advances in deep learning, self supervised representation learning, and neural vocoders. Early speech synthesis systems relied on concatenative and parametric techniques, which produced intelligible but often unnatural and inflexible speech. Modern neural approaches, including sequence-to-sequence models, GAN based vocoders, and diffusion based architectures, have significantly improved speech naturalness and intelligibility [13], [14], [16]. Despite these advances, achieving personalized speech synthesis that preserves speaker identity, expressive prosody, and emotional nuance remains a challenging problem, particularly under data scarcity conditions. Recent state-of-the-art approaches, including diffusion based models and zero-shot voice cloning systems such as YourTTS, have demonstrated high quality synthesis but often come with increased computational complexity and latency, limiting their deployment in real-time applications.

A major research focus has been reducing the amount of speaker specific data required for adaptation while maintaining perceptual quality. Few-shot and zero-shot voice cloning methods attempt to generalize across speakers using shared latent representations and speaker embeddings [12], [20]. At the same time, practical deployment demands lightweight architectures capable of real-time inference, which constrains model complexity and computational requirements [15], [22].

These competing objectives—expressiveness, identity preservation, data efficiency, and deployability, which define the core challenges addressed by contemporary voice cloning research. RVC has emerged as a promising approach in this context. By combining self supervised speech representations, pitch conditioning, and reference-based retrieval, RVC enables effective speaker adaptation using relatively small datasets while avoiding the heavy computational cost of end-to-end text-to-speech retraining [19]. The following subsections review prior work relevant to emotional expressiveness, data efficiency, multilingual adaptation, evaluation standards, and modular system design, which together motivate the design choices of the proposed framework.

Table I provides a list of comparative studies for the voice cloning and personalized speech synthesis in the previous year.

III. EMOTION, PROSODY, AND MODEL EFFICIENCY

Ghadekar et al. [18] proposed an emotionally expressive voice cloning system integrating HuBERT, VQ-VAE, Tacotron2, and recurrent residual networks with reference-based pitch modeling. Their approach demonstrated strong emotional control and high speaker similarity scores, supported by improvements in objective metrics and MOS. However, the multi-stage architecture introduces significant computational overhead and longer training times, limiting real time applicability. Similar trade-offs between expressiveness and efficiency have been reported in multi style voice cloning benchmarks [21].

In contrast, lightweight systems aim to balance expressive capability with computational efficiency. GAN-based vocoders such as Parallel WaveGAN [17] enable fast waveform generation with minimal degradation in quality, making them suitable for real-time pipelines. These findings motivate architectures such as RVC that integrate pitch control and speaker conditioning without relying on large, multi-component model stacks

A. Low-Resource and Cultural Preservation Scenarios

Culafic et al. [26] investigated voice cloning for cultural heritage preservation in low-resource languages by evaluating FastSpeech2, Bark, Coqui TTS, Tortoise, and RVC. Their results demonstrated that RVC performs competitively even with short, carefully curated datasets (approximately 20 minutes), achieving favorable MOS scores relative to more data-intensive models. This study emphasizes the importance of dataset quality over volume, a conclusion echoed in earlier voice conversion challenge [25].

Related work on low-resource adaptation highlights that retrieval-based conditioning and self-supervised representations can substantially reduce reliance on large labeled corpora [23], [24]. These insights directly inform the choice of RVC for personalized voice cloning under constrained data availability.

B. Multilingual and Cross-Lingual Voice Cloning

Multilingual voice cloning aims to maintain a speaker’s identity across different languages by separating linguistic

TABLE I
REPRESENTATIVE RESEARCH ON VOICE CLONING AND PERSONALIZED SPEECH SYNTHESIS

Authors	Year	Dataset	Methodology	Key Results
Jia et al. [1]	2018	LibriSpeech	Speaker embeddings + Tacotron	High quality voice cloning with few seconds of speech
Arik et al. [2]	2018	LibriSpeech	Speaker adaptation in Deep Voice 3	Intelligible cloned speech from limited samples
Qian et al. [3]	2019	VCTK	Autoencoder based voice conversion	Zero-shot voice conversion without parallel data
Ren et al. [4]	2021	LJSpeech, AIShell-3	Variance adaptation with speaker embeddings	Improved speaker similarity with low training data
Casanova et al. [5]	2022	VCTK, LibriTTS	VAE-based speaker modeling	Strong speaker similarity across languages
Kim et al. [6]	2021	LJSpeech	End-to-end VAE + GAN	Natural speech synthesis with fast inference
Min et al. [7]	2021	VCTK	Meta-learning for speaker adaptation	Rapid personalization using few samples
Popov et al. [8]	2021	VCTK	Diffusion probabilistic modeling	High-fidelity voice conversion
So-VITS-SVC Community	2022	VCTK, Custom	VITS-based speaker voice conversion	Lightweight and efficient voice conversion

Seong et al. [20] demonstrated that shared phoneme spaces and speaker embeddings enable cross-lingual synthesis even when training data is monolingual. Likewise principles have been explored in multilingual text-to-speech systems which use universal phoneme representations and autoencoders [28], [29].

Although the primary focus of this study is not on the multilingual synthesis, these results indicate that the RVC-type frameworks, where there is a separation between content and speaker representations, can be scaled to multilingual and code-switched contexts by appropriate data augmentation.

C. Evaluation Standards and Benchmarking

Strict evaluation remains a major challenge in voice cloning studies. The Multi-Speaker Multi-Style Voice Cloning Challenge (M2VoC) [21] established well-known standards based on the MOS, similarity of speakers and similarity of styles of both familiar and unfamiliar speakers. The difficulty of the experiment pointed to the fact that few-shot cloning is often not able to maintain a steady style and identity, stressing the weakness of relying on objective acoustic measurements alone.

Integration of objective measures such as Mel Cepstral Distortion, and cosine similarity with perceptual listening tests has also been emphasized by other studies [27]. These standards provide crucial background to the interpretation of evaluation results in lightweight voice cloning systems.

D. Modular and Deployable Systems

A new system called SynthoTranslate was introduced by Dhabe et al. [22] a modular system that combines speech recognition, translation, text-to-speech synthesis, and lip synchronization for use in accessibility applications. Although their system isn't designed to maintain speaker identity and uses standard TTS voices, it showcases the real-world advantages of modular design, real-time processing, and open-source compatibility. These system-level factors correspond to the design approach of RVC-based pipelines, which focus on flexibility and practical implementation.

IV. DATASET

Our system employs a dual-dataset approach to achieve both broad generalization across various speech patterns and customization to an individual speaker's voice.

VCTK Corpus: Studio-quality English recordings from 109 speakers, including aligned transcriptions and metadata. Each speaker recites around 400 sentences, leading to a total of approximately 44 hours of spoken data. The dataset includes audio recordings sampled at 48 kHz and orthographic transcriptions for every utterance. Metadata including details about the speaker's gender and accent is also provided. It aids in training foundational layers to handle variations in pronunciation and intonation.

Custom Dataset: We collected about 8.5 hours of WAV samples from the target speaker in a quiet environment. This dataset size is adequate for successful speaker adaptation within the RVC framework and corresponds with previous research showing good results even when using minimal data.

V. METHODOLOGY

The architecture of our voice cloning system is designed in a modular and efficient design that enables us to synthesize speech of a high quality with minimum training data. The system breaks down the complex process of voice cloning into four specialized components that undertake important aspects of speech processing. This scalability, easier integration and enhanced performance with other speaker inputs are supported by this modularity. Complete architecture is shown in Fig. 2.

HuBERT Encoder: This component is tasked with extracting linguistic and phonetic data from raw speech inputs. HuBERT (Hidden Unit BERT) uses self-supervised learning on unlabeled audio data, allowing it to be highly effective across various languages and accents. Its self-supervised training allows for language-agnostic adaptation, proving effective even in the absence of transcriptions. By identifying high-level speech components, it minimizes reliance on manually

annotated corpora, making it well-suited for low-resource datasets.

RMVPE: RMVPE (Robust Mel-spectrogram Variance-based Pitch Estimator) is used to extract fundamental frequency (F) contours that define the pitch and prosody of speech. Performs well under noisy or expressive input conditions. Accurate pitch information is essential to maintain natural intonation and speaker-specific cadence, especially for expressive or emotional speech samples. RMVPE ensures reliable F0 estimation, even under challenging audio conditions.

FAISS: Facebook AI Similarity Search (FAISS) is employed to retrieve relevant speaker embeddings from a reference dataset. These embeddings act as a guide to infuse the synthesized voice with the unique vocal characteristics of the target speaker. Maintains identity with low latency and enables generalization to new speakers. Its fast and efficient indexing mechanism ensures scalability when working with large speaker pools.

GAN-based Vocoder: The final waveform is generated with a vocoder that utilizes a Generative Adversarial Network (GAN). This vocoder converts processed linguistic, pitch, and speaker identity embeddings into speech that sounds natural. Exceeds traditional vocoders in maintaining realism and naturalness. The adversarial training framework enables the vocoder to generate high-quality outputs that closely resemble human speech in terms of tone, texture, and dynamics.

A. Training

The training phase includes a comprehensive preprocessing pipeline which breaks the raw audio down into smaller, easier-to-handle bits, removes silence and even normalizes the amplitude to achieve uniformity. The segments are then correlated with their respective phoneme and pitch features respectively using HuBERT and RMVPE. Such representations extracted are relevant to the modeling of the prosodic flow, as well as linguistic structure of the voice of the target speaker. Fine-tuning is the method that is based on speaker-specific recording along with pre-trained RVC weights. The two losses that are combined in the optimization process are the MFCC reconstruction loss and F0 embedding loss. MFCC loss ensures that the speech generated is spectrally similar to the target speech and F0 loss preserves the pitch and prosodic features. This is a two-task optimization that enables the model to best represent the linguistic content as well as the individual vocal traits of the speaker.

B. Inference

In the process of inference, new input speech is passed through two main encoders in the trained model. HuBERT recognizes the linguistic and phonetic features, and RMVPE records the patterns of pitch and intonation. These embeddings are then added with reference vectors of the target speaker that were acquired in the training process. This mixture is passed to the GAN-based vocoder to generate the final audio waveform. Vocoder produces speech which mimics the voice characteristics of the target and maintains the language of the

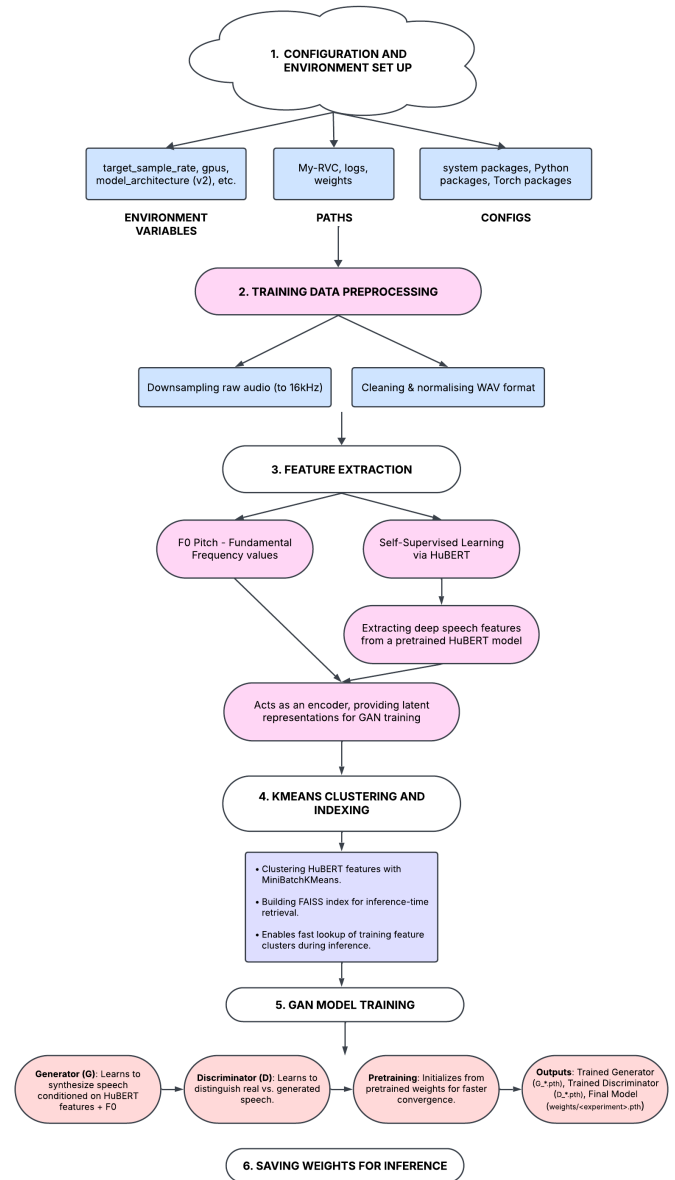


Fig. 2. Deep Learning Pipeline showing the complete workflow from environment setup to model training and saving weights for inference. It includes preprocessing, feature extraction (pitch and HuBERT), clustering, and GAN-based synthesis.

original input. This pipeline supports real-time conversion, and at the same time being very natural, which qualifies it to be used in interactive applications such as in dubbing, assistive speech systems and real-time avatars.

VI. RESULTS

We provide visual analyses of the quality and properties of the cloned speech in terms of waveform, spec-trogram, F0 contour, Mel spectrogram, and chroma. These graphics provide multidimensional information on the naturalness, pitch, tone, and retention of the identity of the speaker in the synthesized voice. All the visual outputs were produced with the help of

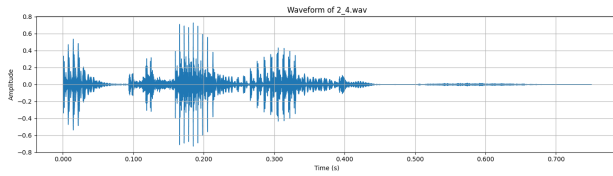


Fig. 3. Waveform of synthesized speech showing smooth amplitude variations, indicating natural speech rhythm and continuity.

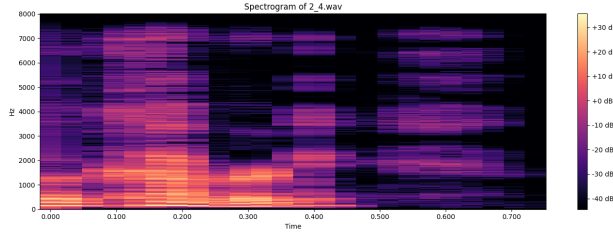


Fig. 4. Spectrogram showing frequency energy distribution over time.

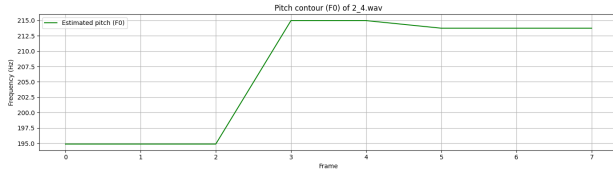


Fig. 5. F0 pitch curve highlighting pitch variation across time frames.

some standard audio processing software and pitch detection methods.

The waveform (Fig. 3) shows that there are obvious changes in the amplitude, which means that voiced and unvoiced speech elements, and silence intervals exist. The lack of disfluency between envelopes implies that the cloned content is able to retain the natural speech rhythm and the prosody that would be very essential in perception and comprehension by the listeners.

Figure 4 illustrates distinct coincidence of spectral characteristics that are typical in natural speech. The formant structures, fricatives and vowel areas are well discernible. The frequencies under 4 kHz are within the range of the human voice and make the speech clearer, and the blurring between the phonemes shows that the pronunciation is clear.

This (Fig. 5) provides the F0 contour, which gives the observation of the intonation and pitch patterns in the cloned speech. There have been natural fluctuations and up-and-down tendencies indicating that emotional expressiveness and tonal fluctuations have been preserved. This validates the pitch modeling component of our architecture and implies that it has a high speaker identity preservation.

As shown in Fig. 6, illustrates that the Mel spectrogram records finer details of the speech in frequency bins which are related to human auditory scales. This includes the variation of tone, the variation in energy between phonemes and the articulation of well-structured formants. The synthesis product has been shown to have a high correspondence to human-

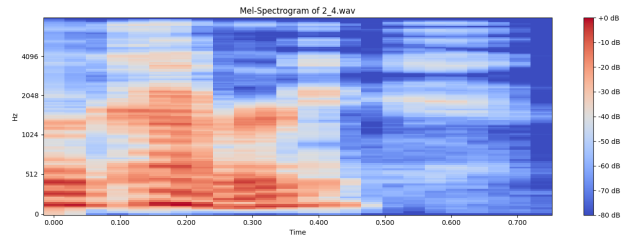


Fig. 6. Mel spectrogram emphasizing perceptually significant frequency bins.

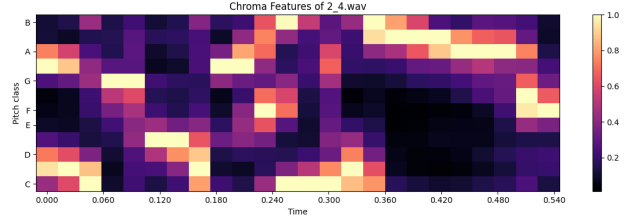


Fig. 7. Chroma features representing pitch class profiles across time.

perceived clarity and phonetic finesse. The chroma plot (Fig. 7) shows the harmonic and pitch class data of the audio signal. It demonstrates tonal steadiness and harmonic consistency particularly in the voiced sections. This indexes that harmonic identity and musical pitch organization is maintained efficiently in the generated voice.

The chroma plot (Fig. 7) represents the harmonic and pitch class information of the audio signal. It shows tonal stability and harmonic consistency, especially in voiced parts. This shows that harmonic identity and musical pitch structure are preserved effectively in the generated voice.

A. Quantitative Speaker Similarity Evaluation

In order to determine the degree of similarity between the synthesized voice and the target speaker, Mel-frequency cepstral coefficient (MFCC) features were derived on five original recordings of the target speaker and the respective synthesized versions. Similarity Cosine similarity was computed between averaged MFCC vectors as shown in Table II.

TABLE II
MFCC COSINE SIMILARITY SCORES

Comparison	Cosine Similarity
Source Speaker vs Target Speaker	0.71
Synthesized Speech vs Target Speaker	0.9935

MFCC cosine similarity is an objective measure that is used to evaluate the acoustic similarity of speech signals. Higher values of similarity imply a more precise preservation of the features of the speaker. Nonetheless, it is not a complete representation of the quality of perceptions and, therefore, qualitative analyses based on spectrograms, pitch contours, and chroma features are also included to make it complete.

The high degree of inter-relationship between source-to-target and synthesized-to-target is indicative of effective speaker identity transfer in the proposed RVC-based model.

Compared to the current few-shot voice cloning systems such as AutoVC and YourTTS, the presented RVC-based system demonstrates competitive results at a significantly lower level of data and computation requirements. The proposed approach has faster inference and lower latency as compared to diffusion-based or large neural TTS systems and is more applicable to real-time and low-resource scenarios.

VII. CONCLUSION AND FUTURE WORK

The present paper presented a lightweight voice cloning system based on RVC that can be adjusted to the vocal characteristics of a target speaker using an insignificant amount of training data. Quantitative evaluation using MFCC cosine similarity and qualitative visual representation indicates that there is successful acoustic alignment between the synthesized and the reference speech. The modular design and real-time inference capabilities of the system make it suitable for practical implementation with few resources. However, the present assessment is restricted to one speaker pair and objective acoustic measurements. Future directions will be to incorporate metrics of perceptual assessment such as MOS and ABX testing, extend the experiments to cover multiple speakers and data sets, and perform against the best voice cloning systems. Furthermore, in order to achieve successful implementation in the real world, it will be necessary to increase resilience in noisy settings and address such ethical concerns as misuse detection and watermarking.

VIII. ETHICAL CONSIDERATIONS AND MISUSE PREVENTION

Voice cloning technologies are a great ethical concern, especially when it comes to abuse of identity, impersonation, and invasion of privacy. In this research, the target speakers were approached with the express agreement and used solely to conduct research. The given system is not supposed to be duplicated by unauthorized voices or used for misleading purposes. Trained models are not publicly released to avoid abuse, and the deployment scenarios make the users provide informed consent. Potential safeguards include the introduction of watermarks to synthesized speech, adoption of disclosure procedures for audio generated, and control of access to model usage. The further study will involve automatic detection of synthesized speech and integration of ethical principles to make sure that personalized voice technologies are used responsibly.

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